

factors are tradeable too. These factors reflect macro risk and at some level should reflect the underlying fundamental risks of the economy. Macro factors like inflation and economic growth, however, are not usually directly traded (at least not in scale, with the exception of volatility), and so dynamic factors have a big advantage that they can be easily implemented in investors' portfolios.

I present examples of dynamic factors using the best-known example of a tradeable multifactor model introduced by Fama and French (1993). I interchangeably use the words "style factors," "investment factors," and "dynamic factors." Sometimes these are also called "smart beta" or "alternative beta," mostly by practitioners.

3.1. FAMA AND FRENCH (1993) MODEL

The Fama–French (1993) model explains asset returns with three factors. There is the traditional CAPM market factor and there are two additional factors to capture a size effect and a value/growth effect:

$$E(r_i) = r_f + \beta_{i,MKT}E(r_m - r_f) + \beta_{i,SMB}E(SMB) + \beta_{i,HML}E(HML), \quad (7.2)$$

where two new factors, *SMB* and *HML*, appear alongside the regular CAPM market factor.

Let us briefly recap the effect of the CAPM market factor (see chapter 6). When the market does poorly, stocks that have high exposures to the market factor (stocks with high betas, $\beta_{i,MKT}$) also tend to do badly. That is, high beta stocks tend to tank in parallel when the market tanks. But over the long run, the CAPM predicts that stocks with high betas will have higher average returns than the market portfolio to compensate investors for losses when bad times hit—defined by the CAPM theory as low returns of the market.

Robert Merton (1973), Stephen Ross (1976), and others developed the theoretical multifactor model framework in the 1970s, but it took another two decades for an explosion of studies to demonstrate that factors other than the market mattered empirically. Two of these effects—size and value—are in the Fama–French model. Fama and French did not discover these effects; they just provided a parsimonious model to capture their effects. Unfortunately for the original authors, most of the credit for these risk factors now gets assigned to Fama and French.

In equation (7.2), the first factor in addition to the market factor in the Fama–French model is *SMB*, which refers to the differential returns of small stocks minus big stocks (hence *SMB*), where small and big refer simply to the market capitalization of the stocks. (Fama and French were clearly not marketers, so the labels on the factors are a tad banal.) The *SMB* factor was designed to capture the outperformance of small firms relative to large firms.

The other factor in the Fama–French model is the *HML* factor, which stands for the returns of a portfolio of high book-to-market stocks minus a portfolio of low book to market stocks. The book-to-market ratio is book value divided by market capitalization, or the inverse of equity value normalized by book value.